SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	P. Chola Lakshmi Narasimha	**Reg. No.:**	192325021
Section / Batch:		**Date:**	

Code of Conduct

I ___________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, wordclass changes, and semantic shifts.	Q4

Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

1. Search Engine Indexing System

Scenario

A search engine processes the words **"analyzing", "analysis", and "analytical"**. To improve search accuracy, the system groups related words under one common representation using morphological analysis.

Step 1: Morphological Decomposition

Word	Root	Affix	Word Class
analyzing	analyze	-ing	Verb (Present Participle)
analysis	analyze	-sis	Noun
analytical	analyze	-ical	Adjective

Step 2: Inflectional vs Derivational

- **analyzing → Inflectional**
  - "-ing" indicates an action in progress.
  - Word class remains a verb.
- **analysis → Derivational**
  - "-sis" converts the verb into a noun.
- **analytical → Derivational**
  - "-ical" converts the base into an adjective.

Step 3: Normalization

All variants are mapped to the common base form:

analyzing
analysis
analytical
↓
analyze

Final Morphological Structure

Input Word	Morphological Structure	Classification	Normalized Form
analyzing	analyze + ing	Inflectional	analyze
analysis	analyze + sis	Derivational	analyze
analytical	analyze + ical	Derivational	analyze

Morphological Processing Pipeline

Input Words
│
▼
Tokenization
│
▼
Identify Prefix/Suffix
│
▼

Extract Root
|
▼
Classify
(Inflectional / Derivational)
|
▼
Normalize to "analyze"
|
▼
Store in Search Index

2. AI-Based Sentiment Analysis

Scenario
The system receives the words:

- disagree
- agreement
- agreeable

It must identify the common base word for consistent sentiment interpretation.

Morphological Parsing
Word 1: disagree
dis + agree
Prefix = dis-
Root = agree

Word 2: agreement
agree + ment
Suffix = -ment
Forms a noun.

Word 3: agreeable
agree + able
Suffix = -able
Forms an adjective.

Classification

Word	Prefix	Root	Suffix	Type
disagree	dis-	agree	—	Derivational
agreement	—	agree	-ment	Derivational
agreeable	—	agree	-able	Derivational

Meaning Change

- **agree** → to have the same opinion
- **disagree** → opposite meaning
- **agreement** → state of agreeing
- **agreeable** → pleasant or willing to agree

Normalized Representation
disagree

agreement
agreeable
↓
agree

Final Output

Word	Morphological Breakdown	Classification	Root
disagree	dis + agree	Derivational	agree
agreement	agree + ment	Derivational	agree
agreeable	agree + able	Derivational	agree

3. News Aggregation System

Scenario
Input Words

- govern
- government
- governance

The system standardizes these words for topic modeling.

Morphological Decomposition
govern
govern
Root word

government
govern + ment
Creates a noun.

governance
govern + ance
Creates a noun.

Derivational Layers

Word	Root	Suffix	Type
govern	govern	—	Root
government	govern	-ment	Derivational
governance	govern	-ance	Derivational

Normalization
govern
government
governance
↓
govern

Final Output

Input	Morphological Structure	Normalized Form
govern	govern	govern
government	govern + ment	govern
governance	govern + ance	govern

Processing Flow
Input
│
▼
Morphological Analysis
│
▼
Extract Root
│
▼
Remove Derivational Suffix
│
▼
Normalize
│
▼
Store "govern"

4. Document Classification System

Scenario
Input Words

- activate
- activation
- reactivation

The system groups related words by identifying prefixes, suffixes, and root forms.

Morphological Parsing
activate
activate
Root word (verb)

activation
activate + ion
Verb → Noun

reactivation
re + activate + ion
Prefix = re-
Suffix = -ion

Morphological Breakdown

Word	Prefix	Root	Suffix	Word Class
activate	—	activate	—	Verb
activation	—	activate	-ion	Noun
reactivation	re-	activate	-ion	Noun

Meaning Changes

- **activate** → make active
- **activation** → process of making active
- **reactivation** → making active again

Classification

Word	Type
activate	Base Form
activation	Derivational
reactivation	Derivational

Unified Representation

activate
activation
reactivation
↓
activate

Final Output

Word	Morphological Structure	Normalized Root
activate	activate	activate
activation	activate + ion	activate
reactivation	re + activate + ion	activate

5. Search Optimization Module

Scenario

Input Words

- create
- creates
- creating

The system removes only inflectional changes for consistent retrieval.

Morphological Decomposition

create
create
Base word

creates
create + s
Third-person singular verb

creating
create + ing
Present participle

Classification

Word	Root	Suffix	Type
create	create	—	Base Form
creates	create	-s	Inflectional
creating	create	-ing	Inflectional

Normalization
create
creates
creating
↓
create

Final Output

Input Word	Morphological Breakdown	Classification	Normalized Form
create	create	Base Form	create
creates	create + s	Inflectional	create
creating	create + ing	Inflectional	create

Morphology-Based Normalization Flow

Input Words
│
▼
Detect Suffix
│
▼
Remove Inflectional Ending
│
▼
Extract Root
│
▼
Normalize to "create"
│
▼
Store in Search Index

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	**Understanding of Scenario**	**Identification of Key Issues**	**Application of Concepts**	**Decision Making**	**Clarity of Response**	**Sub. Total**	**Total %**
1							
2							
3							
4							
5							